

19. (2.5 points)

A company's equity is valued at \$276,716,000 based on the following assumptions:

	2018	2019	2020
Net Income (\$000)	22,000	28,000	34,000
Starting GAAP Equity (\$000)	189,200	190,300	191,700

- The risk free interest rate = 4.0%.
- The equity market risk premium = 5.0%.
- The company's beta = 0.8.
- After 2020, the company's abnormal earnings decrease linearly until reaching zero in 2031.

Financial ratios for several peer companies are summarized below:

	Price/Earnings	Price/Sales	Price/Cash Flow	Price/Book
Peer 1	12.5	1.0	8.0	1.29
Peer 2	13.4	0.9	7.7	1.59
Peer 3	12.5	2.1	14.1	1.37
Peer 4	12.9	4.5	12.8	1.76

a. (1.5 points)

Calculate the change in the estimated equity value if the abnormal earnings are assumed to decrease linearly to zero in 2023 instead of in 2031.

b. (1 point)

Justify whether 2 years or 10 years of non-zero abnormal earnings after 2020 produces a more reasonable answer.

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SPRING 2018 EXAM 7, QUESTION 19	
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVES: B2, B3
SAMPLE ANSWERS	
Part a: 1.5 points	
<u>Sample 1</u> Discount Rate = $4.0\% + 0.80 \times 5.0\% = 8.0\%$ 2018 AE = $22,000 - 189,200 \times 8.0\% = 6,864$ 2019 AE = $28,000 - 190,300 \times 8.0\% = 12,776$ 2020 AE = $34,000 - 191,700 \times 8.0\% = 18,664$ 2021 AE = $18,664 \times 2/3 = 12,443$ 2022 AE = $18,664 \times 1/3 = 6,221$ Equity = $189,200 + 6,864/1.08 + 12,776/1.08^2 + 18,664/1.08^3 + 12,443/1.08^4 + 6,221/1.08^5$ = 234,705 Difference = $234,705 - 276,716 = -42,011$ <u>Sample 2</u> $4.0\% + 0.80 \times 5.0\% = 8.0\%$ 2018 ROE = $22 / 189.2 = 11.6\%$ 2019 ROE = $28 / 190.3 = 14.7\%$ 2020 ROE = $34 / 191.7 = 17.7\%$ $(11.6\% - 8.0\%) \times 189.2 = 6.8$ $(14.7\% - 8.0\%) \times 190.3 = 12.75$ $(17.7\% - 8.0\%) \times 191.7 = 18.6$ $18.6 \times 2/3 = 12.4$ $18.6 \times 1/3 = 6.2$ $189.2 + 6.8/1.08 + 12.75/1.08^2 + 18.6/1.08^3 + 12.4/1.08^4 + 6.2/1.08^5 = 234.5$ $234.5 - 276.7 = -42.2$ <u>Sample 3</u> $4.0\% + 0.80 \times 5.0\% = 8.0\%$ 2020 AE = $34,000 - 191,700 \times 8.0\% = 18,664$ <u>When AE decreases linearly to zero in 2023:</u> 2021 AE = $18,664 \times 2/3 = 12,443$ 2022 AE = $18,664 \times 1/3 = 6,221$	

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Present Value of AE = $12,443/1.08^4 + 6,221/1.08^5 = 13,380$

When AE decreases linearly to zero in 2031:

2021 AE = $18,664 \times 10/11 = 16,967$

2022 AE = $18,664 \times 9/11 = 15,271$

2023 AE = $18,664 \times 8/11 = 13,574$

2024 AE = $18,664 \times 7/11 = 11,877$

2025 AE = $18,664 \times 6/11 = 10,180$

2026 AE = $18,664 \times 5/11 = 8,484$

2027 AE = $18,664 \times 4/11 = 6,787$

2028 AE = $18,664 \times 3/11 = 5,090$

2029 AE = $18,664 \times 2/11 = 3,393$

2030 AE = $18,664 \times 1/11 = 1,697$

PV of AE = $16,967/1.08^4 + \dots + 1,697/1.08^{13} = 55,391$

Difference = $13,380 - 55,391 = -42,011$

Sample 4

$4.0\% + 0.80 \times 5.0\% = 8.0\%$

2020 AE = $34,000 - 191,700 \times 8.0\% = 18,664$

PV of difference in AE's:

$18,664 \times [$

$(2/3 - 10/11) / 1.08^4 +$

$(1/3 - 9/11) / 1.08^5 +$

$(-8/11) / 1.08^6 +$

$\dots +$

$(-1/11) / 1.08^{13}]$

$= -42,011$

Part b: 1 point

Sample 1

Under 2-year assumption:

$P/BV = 234,705 / 189,200 = 1.24$

$P/E = 234,705 / 22,000 = 10.67$

Under 10-year assumption:

$P/BV = 276,716 / 189,200 = 1.46$

$P/E = 276,716 / 22,000 = 12.58$

The 10-year ratios are more in line with peer companies for both P/BV and P/E. Therefore, 10-year is more reasonable.

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Sample 2

Average of peer P/E = $(12.5 + 13.4 + 12.5 + 12.9) / 4 = 12.825$

Estimated equity = $12.825 \times 22,000 = 282,150$

Average of peer P/BV = $(1.29 + 1.59 + 1.37 + 1.76) / 4 = 1.5025$

Estimated Equity = $1.5025 \times 189,200 = 284,273$

These equity values are closer to the equity under the 10-year assumption (276,716) so 10 years is more reasonable.

Sample 3

Peer equities based on P/E:

Smallest Peer P/E: $12.5 \times 22,000 = 275,000$

Largest Peer P/E: $13.4 \times 22,000 = 294,800$

Peer equities based on P/BV:

Smallest Peer P/BV: $1.29 \times 189,200 = 244,068$

Largest Peer P/BV: $1.76 \times 189,200 = 332,992$

The equity under the 2-year assumption is outside these ranges while the equity under the 10-year assumption is within these ranges, so 10-years is more reasonable.

EXAMINER'S REPORT

Candidates were expected to demonstrate understanding of expected abnormal earnings, to use the estimated abnormal earnings to estimate the equity value of a P&C insurer, and to evaluate and justify the reasonableness of different assumptions by using peer company financial ratios.

Part a

Candidates were expected to calculate abnormal earnings and use them to estimate the equity value of a company. There were multiple reasonable approaches to receive full credit when calculating the difference. The most common were:

- Calculate the equity under the “zero-in-2023” scenario, and then take the difference from the equity value provided in the question. This approach usually involved the least amount of calculation for the candidate.
- Calculate the abnormal earnings under both scenarios and then take the difference between the equity values, the present values of all abnormal earnings, or the present values of just the tail values of the two scenarios. All three options would give the same result because the starting book value and the first three abnormal earning values are the same under both scenarios.

Common mistakes included:

- Using an incorrect formula to calculate the abnormal earnings from the net incomes, starting GAAP equities, and the calculated discount rate.
- Using required returns instead of abnormal earnings.
- Incorrectly linearly decreasing the abnormal earnings under either scenario. The most

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common example of this error was to use $\frac{3}{4}$, $\frac{1}{2}$, and $\frac{1}{4}$ of the 2020 abnormal earnings over the next three years, respectively, instead of using $\frac{2}{3}$ and $\frac{1}{3}$ of the 2020 abnormal earnings over the next two years. Under the first set of ratios, the abnormal earnings do not decrease linearly to zero in 2023, as the question states.

- Comparing the undiscounted tail value of the two scenarios.
- Forgetting to take the difference between the equity values of the two scenarios.
- While technically not a mistake, some spent time calculating abnormal earnings that were already provided in the question.

Part b

Candidates were expected to use the peer company ratios provided in the question to determine which of the two different scenarios was more reasonable.

Common mistakes included:

- Using the equity value of 276,716 as the denominator instead of the starting GAAP equity of 189,200.
- Using the book values from 2019 or 2020 instead of 2018.
- Claiming that two years is more reasonable simply because it is fewer than ten years, with no justification for why abnormal earnings are not expected to last forever.